

Supplementary Information

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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All cases, scripts and tabulated results: github.com/Imerson/cryostat-first-shield-He-vs-LN2.

S1 Enclosure model

Table S1. Regions, patches and boundary conditions of the enclosure CHT cases.

Region	Patch	T condition	Radiation condition
domain0 (OVC-shield gap)	warmPlate, OVC	fixed T_{hot}	grey diffuse, $\varepsilon = 0.04$
	shield_slave	fixed T_{cold}	grey diffuse, $\varepsilon = 0.02$
domain1 (inner gap)	warmPlate	fixed T_{hot}	grey diffuse, $\varepsilon = 0.04$
	coldPlate, shield_bottom	fixed T_{cold}	grey diffuse, $\varepsilon = 0.04$
	shield	fixed T_{cold}	grey diffuse, $\varepsilon = 0.02$
	domain1_to_coax_Lk	conjugate (mapped)	grey diffuse, $\varepsilon = 0.02$
coax_L1..L4 (solids)	warm cap / cold cap	fixed T_{hot} / T_{cold}	–
	lateral	conjugate to domain1	–

Table S2. Gas-region settings that make the gaps behave as vacuum. Radiation in OpenFOAM’s `viewFactor` model is solved in fluid regions, so each gap is a fluid region made inert.

Setting	Value
pressure $p = p_{\text{rgh}}$	10^{-3} Pa (<code>pRefValue</code> 10^{-3})
gravity	(0, 0, 0)
transport	constant, $\mu = 10^{-11}$ Pa s, $\text{Pr} = 0.67$
density limits	$\rho_{\text{max}} = 10^{-4}$, $\rho_{\text{min}} = 10^{-12}$ kg m $^{-3}$
radiation solver frequency	every 10 iterations
agglomeration	<code>nFacesInCoarsestLevel</code> 250, <code>featureAngle</code> 20, <code>mergeLevels</code> 1

Table S3. Harness representation. Cable UT-085-SS-SS: 304 stainless outer conductor $A = 1.585 \text{ mm}^2$, PTFE dielectric $A = 2.001 \text{ mm}^2$, 304 stainless centre conductor $A = 0.205 \text{ mm}^2$. Material $k(T)$ from the NIST cryogenic fits $\log_{10} k = \sum_n a_n (\log_{10} T)^n$. Each modelled cylinder carries n_k lines with $\kappa_{\text{eff}}(T) = (n_k/A_{\text{model}}) \sum_i A_i k_i(T)$. A_{model} is the conduction cross-section of the meshed cylinder in each stage span (calibrated from the mesh so that $(A_{\text{model}}/L) \int \kappa_{\text{eff}} dT$ equals the physical cable integral $n_k L^{-1} \sum_i A_i \int k_i dT$); it differs between the stage-1 and stage-2 geometries. Because A_{model} cancels in that product, the stage-integrated conduction is fixed by the cable data by construction: the resolved model reproduces the targets below to 0.01 %, which verifies the implementation but is not an independent prediction.

Span	$A_{\text{model}} [\text{m}^2]$	$L [\text{m}]$	L1 drive, $n_k=25$	L2 flux, 25	L3 readout, 10	L4 pump, 5
300→50 K	2.975×10^{-4}	0.335	395.8 mW	395.8 mW	158.3 mW	79.2 mW
300→77 K	2.976×10^{-4}	0.335	370.0 mW	370.0 mW	148.0 mW	74.0 mW
50→4 K	3.934×10^{-3}	0.245	27.03 mW	27.03 mW	10.81 mW	5.41 mW
77→4 K	2.845×10^{-3}	0.245	62.33 mW	62.33 mW	24.93 mW	12.47 mW

Per-bundle targets $n_k L^{-1} \sum_i A_i \int k_i dT$; shares 38.5 / 38.5 / 15.4 / 7.7 % follow from n_k with identical cable construction and length.

Table S4. Deployed $\kappa_{\text{eff}}(T) = \sum_{n=0}^5 c_n T^n [\text{W m}^{-1} \text{K}^{-1}]$, T in K] per bundle and stage span (`kappaCoeffs<8>`, last two coefficients zero). Fit error < 0.5 % over each span.

Span	Bundle	c_0	c_1	c_2	c_3	c_4	c_5
300→50 K	L1	-2.1522×10^{-1}	3.1627×10^{-2}	-2.3184×10^{-4}	9.9921×10^{-7}	-2.1908×10^{-9}	1.9383×10^{-12}
	L2	-2.1519×10^{-1}	3.1622×10^{-2}	-2.3181×10^{-4}	9.9907×10^{-7}	-2.1905×10^{-9}	1.9380×10^{-12}
	L3	-8.6092×10^{-2}	1.2651×10^{-2}	-9.2741×10^{-5}	3.9970×10^{-7}	-8.7636×10^{-10}	7.7535×10^{-13}
	L4	-4.3044×10^{-2}	6.3253×10^{-3}	-4.6369×10^{-5}	1.9984×10^{-7}	-4.3816×10^{-10}	3.8766×10^{-13}
300→77 K	L1	-1.3459×10^{-1}	2.9103×10^{-2}	-2.0205×10^{-4}	8.3171×10^{-7}	-1.7398×10^{-9}	1.4706×10^{-12}
	L2	-1.3457×10^{-1}	2.9099×10^{-2}	-2.0202×10^{-4}	8.3159×10^{-7}	-1.7396×10^{-9}	1.4704×10^{-12}
	L3	-5.3837×10^{-2}	1.1642×10^{-2}	-8.0824×10^{-5}	3.3269×10^{-7}	-6.9596×10^{-10}	5.8824×10^{-13}
	L4	-2.6917×10^{-2}	5.8206×10^{-3}	-4.0410×10^{-5}	1.6634×10^{-7}	-3.4796×10^{-10}	2.9411×10^{-13}
50→4 K	L1	-4.8624×10^{-4}	9.1378×10^{-4}	3.6621×10^{-5}	-9.0343×10^{-7}	8.4798×10^{-9}	-2.8699×10^{-11}
	L2	-4.8624×10^{-4}	9.1378×10^{-4}	3.6621×10^{-5}	-9.0343×10^{-7}	8.4798×10^{-9}	-2.8699×10^{-11}
	L3	-1.9450×10^{-4}	3.6551×10^{-4}	1.4648×10^{-5}	-3.6137×10^{-7}	3.3919×10^{-9}	-1.1480×10^{-11}
	L4	-9.7247×10^{-5}	1.8275×10^{-4}	7.3242×10^{-6}	-1.8068×10^{-7}	1.6959×10^{-9}	-5.7398×10^{-12}
77→4 K	L1	-7.4665×10^{-4}	1.2814×10^{-3}	4.9472×10^{-5}	-1.2245×10^{-6}	1.1714×10^{-8}	-4.2617×10^{-11}
	L2	-7.4665×10^{-4}	1.2814×10^{-3}	4.9472×10^{-5}	-1.2245×10^{-6}	1.1714×10^{-8}	-4.2617×10^{-11}
	L3	-2.9865×10^{-4}	5.1254×10^{-4}	1.9788×10^{-5}	-4.8978×10^{-7}	4.6855×10^{-9}	-1.7046×10^{-11}
	L4	-1.4932×10^{-4}	2.5626×10^{-4}	9.8938×10^{-6}	-2.4488×10^{-7}	2.3427×10^{-9}	-8.5230×10^{-12}

Free-molecular gas conduction bound. At $p < 10^{-2} \text{ Pa}$ the mean free path exceeds the gap and gas heat transport is free-molecular, $\dot{Q}_{\text{gas}} = \alpha \Lambda p A \Delta T$ [1], with $\Lambda = \frac{\gamma+1}{\gamma-1} \sqrt{R/(8\pi MT)}$ evaluated at the warm wall (helium, $\gamma = 5/3$: $\Lambda = 2.10 \text{ W m}^{-2} \text{K}^{-1} \text{Pa}^{-1}$; nitrogen: 1.19), accommodation coefficient α , exchange area A and $\Delta T = 250 \text{ K}$. For the enclosure's equivalent plate area $A = 1.14 \text{ m}^2$ and helium with $\alpha = 0.4$ the bound is 2.4 mW at 10^{-5} Pa , 24 mW at 10^{-4} Pa and 0.24 W at 10^{-3} Pa , that is 0.2, 2.3 and 23 % of the 1.03 W harness conduction ($\alpha = 1$: 0.6, 5.8 and 58 %). The model therefore applies to a cryostat pumped to 10^{-4} Pa or better, where the residual gas is a budget line of a few per cent; the nominal $p = 10^{-3} \text{ Pa}$ of the gap regions is a solver setting for the radiation-only vacuum, not an operating pressure.

Table S5. View-factor agglomeration refinement and row normalisation. Closure = mean row sum $\sum_j F_{ij}$ of the generated matrix (unity for a closed enclosure); loads with the matrix as generated and, at the production level, with rows normalised.

Case	Coarsest-level faces	Closure (mean; range)	Radiation, as generated		Radiation, normalised
300→50 K	60	–	9.46 W		–
300→50 K	250 (production)	0.99–1.01; 0.50–1.13	8.41 W		7.36 W
300→50 K	400	0.98–0.99	6.49 W		7.32 W
300→50 K	800	0.90–0.96	3.26 W		7.27 W
300→50 K	closed form	–	7.07 W		
300→77 K	250 (production)	0.99	8.38 W	7.34 W (closed form 7.04)	
50→4 K	250 (production)	1.00	5.29 mW	3.16 mW (closed form 3.15)	
77→4 K	250 (production)	1.00	29.7 mW	17.8 mW (closed form 17.7)	

Row normalisation. Each fluid region of the enclosure model is a closed surface, so every row of the view-factor matrix must sum to unity. The generated matrix (`viewFactorsGen`, Gauss quadrature tolerance 0.01) is stored sparsely, one entry per visible face pair; the normalisation divides each row by its sum and leaves rows with a sum below 0.2 (none occurred) untouched. The solve was then restarted from the converged production field for 600 further iterations. Script `normalise_F.py` in the repository.

Table S6. Emissivity sensitivity (production view factors) and MLI-equivalent variant; the MLI-equivalent cases are also given with the row-normalised matrix, which is the value carried into the main text and the ledger.

Case	ε warm / cold	Radiation [W]	Total [W]
300→50 K, low	0.02 / 0.01	5.45	6.48
300→50 K, baseline	0.04 / 0.02	8.41	9.44
300→50 K, high	0.06 / 0.03	11.48	12.51
300→77 K, low	0.02 / 0.01	5.44	6.40
300→77 K, baseline	0.04 / 0.02	8.38	9.35
300→77 K, high	0.06 / 0.03	11.44	12.40
300→50 K, MLI-eq.	0.04 / 0.003	1.81	2.84
300→77 K, MLI-eq.	0.04 / 0.003	1.81	2.77
300→50 K, MLI-eq., rows normalised	0.04 / 0.003	1.69	2.72
300→77 K, MLI-eq., rows normalised	0.04 / 0.003	1.68	2.65

S2 Cold-plate sweep: full results

Table S7. All cold-plate runs at 9.4 W (production mesh unless marked). $\dot{m}$ in g s^{-1} , volumetric flow in $\text{cm}^3 \text{s}^{-1}$, Δp in Pa, $\dot{W}_p$ in μW , temperatures in K, h in $\text{W m}^{-2}\text{K}^{-1}$; $T_{\text{in}} = 45 \text{ K}$ (He), 77 K (LN_2). T_w is the area mean over the heated-wall patch, $T_{w,\text{max}}$ and $T_{w,\text{min}}$ its extreme face values (`surfaceFieldValue` on the patch); $\text{margin} = T_{\text{sat}}(5 \text{ bar}) - T_{w,\text{max}} = 94.0 \text{ K} - T_{w,\text{max}}$; $y^+ =$ maximum on the heated wall; $\text{closure} =$ energy-balance error. Gravity-on (g on) nitrogen runs did not reach a steady state (see closure).

Case	Re	$\dot{m}$	$\dot{V}$	Δp	$\dot{W}_p$	T_{out}	T_w	$T_{w,\text{max}}$	$T_{w,\text{min}}$	$T_w - T_b$	h	Nu	f	NTU	ε	Gr/ Re^2	margin	y^+	closure %
He 1 bar $Re=500$	500	0.159	167	0.294	48.6	56.36	87.4	106.1	47.9	36.72	17.1	3.66	0.186	0.31	0.266	0.63	—	—	-0.06
He 1 bar $Re=500$ (g on)	500	0.159	167	0.331	54.6	56.21	67.5	95.2	47.9	16.87	37.1	7.96	0.21	0.674	0.49	0.29	—	—	-1.41
He 1 bar $Re=500$ (variable properties)	500	0.159	165	0.309	51.1	56.37	83.0	97.3	48.0	32.29	19.4	4.16	0.196	0.352	0.297	0.63	—	—	-0.01
He 1 bar $Re=1000$	1000	0.318	316	0.671	211	50.68	76.8	92.6	47.1	28.94	21.7	4.64	0.112	0.196	0.178	0.15	—	—	-0.05
He 1 bar $Re=1500$	1500	0.477	465	1.12	521	48.79	71.7	85.9	46.8	24.77	25.3	5.42	0.0847	0.153	0.142	0.06	—	—	-0.05
He 1 bar $Re=2300$	2300	0.731	702	1.98	1.39e+03	47.47	67.1	79.7	46.6	20.82	30.1	6.45	0.0642	0.119	0.112	0.022	—	—	-0.06
He 1 bar $Re=2300$ (g on)	2300	0.731	703	2.02	1.42e+03	47.48	62.7	80.9	46.6	16.41	38.2	8.18	0.0656	0.151	0.14	0.018	—	—	0.55
He 1 bar $Re=2300$ (coarse grid)	2300	0.731	702	1.95	1.37e+03	47.47	67.1	79.9	47.3	20.85	30.1	6.44	0.0632	0.119	0.112	0.022	—	—	-0.03
He 1 bar $Re=2300$ (fine grid)	2300	0.731	702	1.99	1.4e+03	47.47	67.0	79.7	46.2	20.79	30.1	6.46	0.0647	0.119	0.112	0.022	—	—	-0.07
He 1 bar $Re=2300$ (variable properties)	2300	0.731	702	1.98	1.39e+03	47.47	66.0	76.9	46.7	19.80	31.7	6.78	0.0643	0.125	0.117	0.022	—	—	-0.01
He 1 bar $Re=5000$	5000	1.59	1.5e+03	6.62	9.95e+03	46.14	55.3	61.3	46.5	9.73	64.4	13.8	0.0461	0.117	0.11	0.0023	—	2.9	-0.08
He 1 bar $Re=10000$	10000	3.18	2.99e+03	21.8	6.51e+04	45.57	50.3	54.3	46.4	5.00	125	26.8	0.0382	0.114	0.107	0.0003	—	5.6	-0.18
He 5 bar $Re=2300$	2300	0.731	140	0.395	55.5	47.47	67.1	79.8	46.6	20.82	30.1	6.45	0.0641	0.119	0.112	0.56	—	—	-0.04
He 18 bar $Re=2300$	2300	0.731	39	0.11	4.29	47.47	67.1	79.7	46.6	20.82	30.1	6.45	0.0643	0.119	0.112	7.3	—	—	-0.04
LN_2 3 bar $Re=500$	500	4.09	5.06	0.202	1.02	78.13	85.0	89.2	77.7	7.46	84	5.77	0.163	0.151	0.14	40	4.8	—	0.00
LN_2 3 bar $Re=500$ (g on)	500	4.09	5.06	1.68	8.5	77.32	78.9	81.7	77.7	1.74	361	24.8	1.35	0.65	0.478	9.4	12.3	—	-71.96
LN_2 3 bar $Re=1000$	1000	8.17	10.1	0.501	5.07	77.56	83.1	86.6	77.6	5.77	109	7.45	0.101	0.0977	0.0931	7.8	7.4	—	0.00
LN_2 3 bar $Re=1500$	1500	12.3	15.2	0.869	13.2	77.38	82.1	85.3	77.5	4.92	127	8.75	0.0778	0.0765	0.0736	2.9	8.7	—	0.00
LN_2 3 bar $Re=2300$	2300	18.8	23.3	1.57	36.6	77.25	81.2	84.1	77.5	4.12	152	10.4	0.0599	0.0595	0.0578	1	9.9	—	-0.00
LN_2 3 bar $Re=2300$ (g on)	2300	18.8	23.3	3	69.8	77.22	78.5	79.9	77.5	1.35	463	31.8	0.114	0.181	0.166	0.34	14.1	—	-10.77
LN_2 3 bar $Re=2300$ (coarse grid)	2300	18.8	23.3	1.55	36.1	77.25	81.2	84.1	77.7	4.09	153	10.5	0.0592	0.06	0.0582	1	9.9	—	0.01
LN_2 3 bar $Re=2300$ (fine grid)	2300	18.8	23.3	1.58	36.8	77.25	81.3	84.1	77.4	4.14	151	10.4	0.0603	0.0592	0.0575	1.1	9.9	—	-0.02
LN_2 3 bar $Re=5000$	5000	40.9	50.6	5.57	282	77.11	78.6	80.0	77.4	1.55	405	27.8	0.0449	0.073	0.0704	0.083	14.0	2.9	0.04
LN_2 3 bar $Re=10000$	10000	81.7	101	18.6	1.88e+03	77.06	77.8	78.6	77.4	0.81	777	53.3	0.0374	0.0699	0.0675	0.011	15.4	5.6	0.03

Table S8. Grid convergence index study for both fluids at $Re = 2300$ (Celik et al. 2008); meshes 86 400 / 291 600 / 984 150 cells, refinement ratio 1.5, production mesh = medium.

Fluid	Quantity	Coarse	Medium	Fine	p	Richardson	GCI_{medium}	GCI_{fine}
He, 1 bar	Nu	6.439	6.449	6.457	0.64	6.485	0.70 %	0.54 %
He, 1 bar	f	0.0632	0.0642	0.0647	1.40	0.0655	2.5 %	1.4 %
He, 1 bar	h [$\text{W m}^{-2}\text{K}^{-1}$]	30.05	30.10	30.14	0.64	30.27	0.70 %	0.54 %
LN ₂ , 3 bar	Nu	10.513	10.436	10.382	0.87	10.26	2.2 %	1.5 %
LN ₂ , 3 bar	f	0.0592	0.0599	0.0603	1.71	0.0607	1.6 %	0.8 %
LN ₂ , 3 bar	h [$\text{W m}^{-2}\text{K}^{-1}$]	153.2	152.1	151.3	0.87	149.5	2.2 %	1.5 %

Property correction and energy accounting. The helium transport properties of the earlier version of this comparison ($\mu = 2.73 \mu\text{Pa s}$, $k = 0.0185 \text{ W m}^{-1}\text{K}^{-1}$) correspond to helium near 15 K; the present runs use the 50 K values ($6.36 \mu\text{Pa s}$, $0.0467 \text{ W m}^{-1}\text{K}^{-1}$, CoolProp/NIST). The mixing-cup outlet temperature is the mass-flux-weighted average $\sum \phi_f T_f / \sum \phi_f$ over the outlet faces (`surfaceFieldValue`, `weightedAverage` with `phi`); the `weightedAreaAverage` operation, which multiplies the mass flux by the face area a second time, biases the outlet temperature low on a graded mesh and was the cause of the 2–10 % energy imbalances reported earlier. The delivered wall heat flux was verified with the solver-native `wallHeatFlux` function object (9.395 W).

Wall temperatures and the nitrogen loop pressure. Wall temperatures are read from the heated-wall patch with `surfaceFieldValue` (`max`, `min`, `areaAverage`; include file `system/wallTstats` in every case), and the wall-profile figure of the manuscript uses the patch face values along the centre row of the first channel (`postProcessing/wallFaces/wall_centreline.csv`). A line sample 0.4 mm inside the fluid, which an earlier post-processing used, underestimates the hottest wall face by 8–10 K for helium and 1–3.5 K for nitrogen and is not used anywhere in the paper. The nitrogen cases are stored under the name `LN2_3bar_*` because their properties were generated at 3 bar; the Boussinesq fields do not depend on the loop pressure (density, viscosity and conductivity of the 77 K liquid change by 0.06, 0.3 and 0.1 % between 3 and 5 bar), and the saturation margins of Table S7 are evaluated at the 5 bar loop pressure of the manuscript, with the 3 bar value (87.9 K) quoted where it matters.

S3 Hardware efficiencies used in the real ledger

Table S9. Stage efficiencies and the nitrogen supply-energy multiplier used in the real ledger.

Stage	η range	Source
4 K cryocooler stage	0.008–0.015	PT415: 1.5 W at 4.2 K with 40 W at 45 K for 10.7 kW [2]; the two stages together return 0.031 of their combined Carnot work, which the per-stage ranges reproduce to within 7 % at their midpoints; Radebaugh: “at 4 K efficiencies of about 1 % of Carnot or less are typical” [3]
He first stage (45–80 K)	0.10–0.15	Radebaugh: 10–15 % of Carnot for the best commercial 80 K coolers; large plants up to 30 % [3, 4]
LN ₂ supply (basis of Table 6)	$M_{N_2} = 9.1\text{--}12.7$ W per W intercepted	plant consumption 0.5–0.7 kWh kg ^{−1} [5] divided by the latent heat 199 kJ kg ^{−1} at 1 bar; equivalent to 0.30–0.40 of Carnot at 77 K (ideal liquefaction work 0.213 kWh kg ^{−1} [6]); the fraction-of-Carnot form gives 7.2–9.7 W/W and changes the nitrogen totals by 1–2 %

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